

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: MLA03

COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO3: *Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3, BL6).*

Assessment Tool Weightage: 25%

Dr G. A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

Data Interpretation

Duration: 1hr

Code of Conduct:

I U. Somesh Kumar (192425019) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: U. Somesh Kumar

1)An epsilon-greedy agent is tested with $\epsilon = 0.1, 0.3$, and 0.5 . The cumulative reward is highest at $\epsilon = 0.1$ and lowest at $\epsilon = 0.5$.

RL Concept

Exploration vs. Exploitation

Given Data

Epsilon (ϵ)	Observation
0.1	Highest cumulative reward
0.3	Moderate cumulative reward
0.5	Lowest cumulative reward

Analysis

The ϵ -greedy strategy balances **exploration** (trying new actions) and **exploitation** (choosing the best-known action).

- **$\epsilon = 0.1$:** The agent exploits learned knowledge most of the time and explores only occasionally. This results in higher cumulative rewards because it frequently selects the optimal action.
- **$\epsilon = 0.3$:** The agent explores more often, reducing the number of optimal actions selected and slightly lowering the reward.
- **$\epsilon = 0.5$:** The agent spends half of its decisions exploring, which leads to many non-optimal actions and the lowest cumulative reward.

Interpretation

A lower ϵ value provides a better balance after sufficient learning because the agent can exploit its learned policy while still exploring occasionally to avoid local optima.

Conclusion

Among the tested values, **$\epsilon = 0.1$** provides the best performance because it achieves the highest cumulative reward by effectively balancing exploration and exploitation.

2)Q-Learning and SARSA are compared over 100, 200, and 300 episodes. Q-Learning consistently achieves slightly higher rewards.

RL Concept: Comparison of Q-Learning and SARSA

Given Data

Algorithm	Observation
Q-Learning	Higher average reward
SARSA	Slightly lower average reward

Analysis

- Q-Learning is an off-policy algorithm. It updates Q-values using the maximum possible future reward, allowing it to learn the optimal policy more aggressively.
- SARSA is an on-policy algorithm. It updates values based on the action actually taken, making learning more conservative.

Since Q-Learning consistently achieves higher rewards, it indicates faster convergence toward the optimal policy.

Interpretation

Q-Learning performs better because it always considers the best possible future action during learning, while SARSA incorporates exploratory actions into its updates.

Conclusion

Q-Learning converges faster and achieves higher rewards than SARSA, making it more suitable when the objective is rapid learning of an optimal policy.

3)An RL agent is trained using discount factors $\gamma = 0.2, 0.5$, and 0.9 . The average reward increases as γ increases.

RL Concept

Discount Factor (γ)

Given Data

Discount Factor (γ)	Observation
0.2	Lower reward
0.5	Moderate reward
0.9	Highest reward

Analysis

The discount factor determines how much the agent values future rewards.

- $\gamma = 0.2$: The agent focuses mainly on immediate rewards, which may ignore beneficial long-term strategies.
- $\gamma = 0.5$: The agent balances immediate and future rewards.
- $\gamma = 0.9$: The agent strongly considers future rewards, enabling better long-term decision-making and higher cumulative rewards.

Interpretation

As γ increases, the agent learns policies that maximize long-term returns instead of short-term gains, leading to improved performance.

Conclusion

A discount factor of 0.9 is the most suitable because it provides the highest average reward and supports effective long-term reward optimization.

4)An RL agent initially takes 50 steps to reach the goal, but after 300 episodes it reduces the number of steps to 15.

RL Concept

Policy Improvement and Convergence

Given Data

Training Stage	Steps to Goal
Initial	50 steps
After 300 Episodes	15 steps

Analysis

The reduction in the number of steps indicates that the agent is learning a more efficient policy.

- Initially, the agent explores different paths, resulting in longer routes.
- With more training episodes, the agent gains experience and identifies shorter, more effective paths to the goal.

A decrease from 50 to 15 steps represents a significant improvement in learning efficiency.

Interpretation

The agent has successfully improved its policy by minimizing unnecessary actions and moving closer to the optimal solution.

Conclusion

The reduction in steps strongly suggests that the agent is converging toward an optimal policy, demonstrating successful learning and improved decision-making.

5)An RL agent trained in a stochastic environment shows fluctuating rewards in early episodes but stable rewards after 500 episodes.

RL Concept: Learning Stability and Policy Convergence

Given Data

Training Phase	Reward Trend
Early Episodes	Large fluctuations
After 500 Episodes	Stable rewards

Analysis

In the early stages, the agent explores different actions, leading to inconsistent rewards due to uncertainty and the stochastic nature of the environment.

As training progresses:

- The agent gains experience.
- The policy improves through repeated updates.
- Rewards become more consistent.
- Variance in performance decreases.

Stable rewards indicate that the learned policy performs reliably across different situations.

Interpretation

The transition from fluctuating to stable rewards demonstrates that the agent has developed a robust policy capable of handling uncertainty while maintaining consistent performance.

Conclusion

The stabilization of rewards after 500 episodes indicates successful policy convergence and robust learning, showing that the agent has learned an effective strategy for operating in a stochastic environment.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand